Elliptic PDEs

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Lecture 1

0 Introduction

We will study second order elliptic PDEs on (a domain in) $\mathbb{R}^n$. For example, arising from variational problems. Ultimately, we are interested in non-linear PDEs. To do this, we will first understand the linear theory.

Setup: Consider a domain $\Omega \subseteq \mathbb{R}^n$, i.e. open, bounded and connected, and a function

$$\begin{aligned} F : \Omega \times \mathbb{R} \times \mathbb{R}^n &\to \mathbb{R} \\ (x, z, p) &\mapsto F(x, z, p) \end{aligned}$$

and consider the functional

$$\mathcal{F}[u] = \int_\Omega F(x, u(x), \partial u(x)) \mathrm{d}x$$

Note we will use ∂, D, ∇ essentially interchangably. Assume F is sufficiently regular. Let $u \in \mathcal{S}$, a suitable vector space of functions $u : \Omega \to \mathbb{R}$. Frequently, $\mathcal{S} = \mathrm{H}^1(\Omega)$, or $\mathcal{S} = C^{1,\alpha}(\Omega)$.

Suppose u minimises $\mathcal{F}$, subject to $u|_{\partial\Omega} = g$, for some given $g : \partial\Omega \to \mathbb{R}$[1]. So for all $\varphi \in \mathcal{S}$,

$$\mathcal{F}[u + t\varphi] \geq \mathcal{F}[u]$$

In particular, this means that

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathcal{F}[u + t\varphi] = 0$$

Or another words,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \int_\Omega F(x, u + t\varphi, \partial u + t\partial\varphi) \mathrm{d}x = 0$$

Assume enough regularity so that we can exchange the derivative and integral, we get that

$$\int_\Omega (\partial_z F)(x, u, \partial u)\varphi + \partial_i \varphi (\partial_{p_i} F)(x, u, \partial u) \mathrm{d}x = 0 \tag{1}$$

As usual, we will use the summation convention. To ensure that $u + t\varphi$ satisfies the correct boundary conditions, $\varphi|_{\partial\Omega} = 0$. Integrate eq. (1) by parts, we get

$$\int_\Omega \varphi(x) \left(\partial_z F - \partial_i \partial_{p_i} \mathcal{F}\right)(x, u, \partial u) \mathrm{d}x = 0$$

*Based on lectures by Greg Taujunskas (lectures 1 - 12) and Neshan Wickramasekera (lectures 13 - 24). Last updated January 24, 2024.

[1] Boundary conditions are needed for well-posedness

This is true for all $\varphi \in \mathcal{S}$, and so by the fundamental lemma of calculus of variations,

$$\frac{\partial F}{\partial z} - \frac{\partial}{\partial x_i}\left(\frac{\partial F}{\partial p_i}\right) = 0$$

which is the Euler-Lagrange equation for F. We can rewrite this as

$$\frac{\partial F}{\partial z} - \partial_i \partial_j u \frac{\partial^2 F}{\partial p_i \partial p_j} = 0 \tag{2}$$

This is now a second order quasilinear PDE in u. More generally, consider

$$a^{ij}(x, u, \partial u)\partial^2_{ij} u - b(x, u, \partial u) = 0 \tag{3}$$

Definition 0.1
We say that eq. (3) is *elliptic* in Ω if $a^{ij}(x, u, \partial u)$ is a positive definite matrix in Ω.

In the case of eq. (2), this is equivalent to F being convex in the variable p.

Example 0.2 (Dirichlet energy)
When

$$F(x, z, p) = |p|^2$$

One gets

$$\Delta u = 0$$

Extremisers of this are called *harmonic functions*.

Example 0.3 (Minimal surfaces)
When

$$F(x, z, p) = \sqrt{1 + |p|^2}$$

We leave as an exercise to interpret $\mathcal{F}[u]$. In this case, we get the *minimal surface equation*, which is

$$\nabla \cdot \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0 \tag{4}$$

Locally, ∇u is a constant, and so eq. (4) looks like $\Delta u = 0$, and so solutions have similar local properties. But the existence theory for Laplace's equation is 'trivial', whereas the existence theory for eq. (4) may fail. That is, global properties are important.

For entire solutions (i.e. solutions defined all of $\mathbb{R}^n$), global behaviour is very different.

Theorem 0.4 (Liouville). If $u : \mathbb{R}^n \to \mathbb{R}$ is C^2, $\Delta u = 0$ and u is bounded, then u is a constant.

Theorem 0.5 (Bernstein). (The only entire solutions to eq. (4) in $\mathbb{R}^n$ are planar (i.e. u is linear)) if and only if $n \leq 7$.

1 Harmonic functions

1.1 Basic properties

Let $\Omega \subseteq \mathbb{R}^n$ be a domain, i.e. open and connected.

Definition 1.1 (hamonic, subharmonic, superharmonic)
A function $u \in C^2(\Omega)$ is *harmonic* if $\Delta u = 0$. It is *subharmonic* if $\Delta u \geq 0$ and *superharmonic* if $\Delta u \leq 0$.

Let $B_\rho(y)$ denote the open ball with centre y and radius ρ. Then

Theorem 1.2 (mean value property). If $u \in C^2(\Omega)$ is subharmonic, and $B_r(y) \subseteq \Omega$, then

$$u(y) \leq \frac{1}{\omega_n r^n} \int_{B_r(y)} u(x)\mathrm{d}x \tag{5}$$

where $\omega_n = |B_1(0)|$. Moreover,

$$u(y) \leq \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(y)} u(x)\mathrm{d}x \tag{6}$$

If u is superharmonic, then the inequalities are reversed. If u is harmonic, then equality holds.

Proof. We have that

$$\begin{aligned} 0 &\leq \int_{B_\rho(y)} \Delta u \mathrm{d}x \\ \text{integrating by parts} &= \int_{\partial B_\rho(y)} \nabla u \cdot w \mathrm{d}x \text{ where } w \text{ is the outwards normal} \\ &= \rho^{n-1} \int_{S^{n-1}} w \cdot \nabla u(y + \rho w)\mathrm{d}w \\ &= \rho^{n-1} \int_{S^{n-1}} \frac{\partial}{\partial \rho} u(y + \rho w)\mathrm{d}w \end{aligned}$$

where we use the fact that $\rho w = x - y$. Exchanging integrals and derivatives,

$$0 \leq \frac{\partial}{\partial \rho} \int_{S^{n-1}} u(y + \rho w)\mathrm{d}w$$

Thus, the map

$$\rho \mapsto \int_{S^{n-1}} u(y + \rho w)\mathrm{d}w$$

is increasing. Thus,

$$\int_{S^{n-1}} u(y + \rho w)\mathrm{d}w \leq \int_{S^{n-1}} y(y + rw)\mathrm{d}w$$

for $0 \leq \rho \leq r$. Taking the limit as $\rho \to 0$, we get eq. (6). Integrating in r to get eq. (5). The superharmonic case is similar. The harmonic case follows from the subharmonic and superharmonic cases. □

Remark 1.3. The mean value property characterises harmonic functions. See examples sheet 1.

Lecture 2

Theorem 1.4 (strong maximum principle). Suppose $u \in C^2(\Omega)$ is subharmonic on Ω. Suppose there exists $y_0 \in \Omega$ such that

$$u(y_0) = \sup_\Omega u$$

Then u is constant.

Remark 1.5. If u is superharmonic, then we have a corresponding statement for when $u(y_0) = \inf_\Omega u$. If u is harmonic, then either sup or inf work.

Proof. Let $M = \sup_\Omega u < \infty$, and

$$\Sigma = \{y \in \Omega \mid u(y) = M\}$$

By assumption, Σ is non-empty, as $y_0 \in \Sigma$. Since u is continuous, Σ is closed. Since Ω is connected, it suffices to show that Σ is open, since this implies $\Sigma = \Omega$.

Pick $y \in \Sigma$. By the mean value property, for $\rho > 0$ such that $\overline{B_\rho(y)} \subseteq \Omega$, we have that

$$M = u(y) \leq \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} u(x) \mathrm{d}x$$

So

$$\int_{B_\rho(y)} (M - u(x)) \mathrm{d}x \leq 0$$

But $M - u(x) \geq 0$, and so it must be identically zero, i.e. $u \equiv M$ on $B_\rho(y)$. Hence $B_\rho(y) \subseteq \Sigma$, and Σ is open. □

Here, the strong maximum principle is easy as we have the mean value property. For more general elliptic PDEs, this is not the case. We will prove a weaker statement which does generalise.

Theorem 1.6 (weak maximum principle)**.** Suppose $\Omega \subseteq \mathbb{R}^n$ is a bounded domain, and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$. If u is subharmonic on Ω, then

$$\sup_\Omega u = \sup_{\partial\Omega} u$$

Proof. This follows from the strong maximum principle. Since Ω is bounded, $\sup_\Omega u$ is attained in $\overline{\Omega}$. So by the maximum principle, these cannot be attained in Ω unless u is constant. □

Remark 1.7. If u is superharmonic, we replace sup with inf, and if u is harmonic then both hold.

The mean value property states that u always an average of itself. In particular, this suggest that u cannot vary too much. Can we use this to relate $\sup u$ and $\inf u$?

Theorem 1.8 (Harnack's inequality)**.** Suppose $u \in C^2(\Omega)$, $u \geq 0$ and $\Delta u = 0$ in Ω. Then if $\Omega' \Subset \Omega$ is any bounded subdomain, we have

$$\sup_{\Omega'} u \leq C \inf_{\Omega'} u$$

where $C = C(n, \Omega, \Omega')$ does not depend on u.

Proof. First, choose $y \in \Omega$ and $\rho > 0$, such that $\overline{B_{4\rho}(y)} \subseteq \Omega$. Choose $x_1, x_2 \in B_\rho(y)$. By the mean value property,

$$u(x_1) = \frac{1}{\omega_n \rho^n} \int_{B_\rho(x_1)} u \leq \frac{1}{\omega_n \rho^n} \int_{B_{2\rho}(y)} u$$

On the other hand,

$$u(x_2) = \frac{1}{\omega_n (3\rho)^n} \int_{B_{3\rho}(y)} u \geq \frac{1}{\omega_n (3\rho)^n} \int_{B_{2\rho(y)}} u$$

Combining these, we see that

$$u(x_1) \leq 3^n u(x_2)$$

for all $x_1, x_2 \in B_\rho(y)$. So Harnack holds locally in balls, with constant independent of u. It is also independent of ρ, y as long as ρ is sufficiently small. Now choose x_1, x_2 in $\overline{\Omega'} \subseteq \Omega$, such that

$$\sup_{\Omega'} u = u(x_1) \quad \text{and} \quad \inf_{\Omega'} u = u(x_2)$$

By path connectedness of Ω, there exists a continuous map $\gamma : [0,1] \subseteq \overline{\Omega'}$, with $\gamma(0) = x_1$ and $\gamma(1) = x_2$. Choose ρ such that $4\rho < d(\gamma, \partial\Omega)$, and $N = N(\Omega', \Omega)$ such that we can cover γ by N-balls of radius ρ[2].

Apply the local result to each ball, and we get

$$u(x_1) \leq (3^n)^N u(x_2) = 3^{nN} u(x_2)$$

□

Theorem 1.9 (derivative estimates). Suppose $u \in C^3(\Omega)$ is harmonic on Ω. Then if $\overline{B_\rho(y)} \subseteq \Omega$, we have that

$$|\mathrm{D}u(y)| \leq \frac{C}{\rho} \sup_{\partial B_\rho(y)} |u|$$

where $C = C(n)$.

Proof. Since $\Delta u = 0$,

$$0 = \mathrm{D}_i(\Delta u) = \Delta(\mathrm{D}_i u)$$

So $\mathrm{D}_i u$ is harmonic. By the mean value property,

$$\begin{aligned}
\mathrm{D}_i u(y) &= \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} u \\
&= \frac{1}{\omega_n \rho^n} \int_{B_\rho(y)} \nabla \cdot (0, \dots, u, \dots, 0) \mathrm{d}x \\
&= \frac{1}{\omega_n \rho^n} \int_{\partial B_\rho(y)} u(x) \cdot \nu_i(x) \mathrm{d}x
\end{aligned}$$

where $\nu(x)$ is the unit normal at x. But $|\nu_i(x)| \leq 1$, and so

$$|\mathrm{D}_i u(y)| \leq \frac{1}{\omega_n \rho^n} \sup_{\partial B_\rho(y)} |u| \int_{\partial B_\rho(y)} \mathrm{d}x = \frac{n}{\rho} \sup_{\partial B_\rho(y)} |u|$$

□

Remark 1.10. We can apply this result repeatedly, to get that for $\Omega' \subseteq \Omega'' \subseteq \Omega$, and any multi-index α, if $u \in C^{|\alpha|+2}(\Omega)$, with $\Delta u = 0$ in Ω, then

$$\sup_{\Omega''} |\mathrm{D}^\alpha u| \leq C \sup_{\Omega''} |u|$$

for some $C = C(n, \alpha, \Omega, \Omega')$.

That is,

$$\left\| \mathrm{D}^\alpha u \right\|_{L^\infty(\Omega')} \leq C \| u \|_{L^\infty(\Omega')}$$

By the mean value property, for some $y \in \overline{\Omega''} \subseteq \Omega$,

$$\sup_{\Omega''} u = |u(y)| = \left| c \int_{B_\rho(y)} u(x) \mathrm{d}x \right| \leq C \int_\Omega |u|$$

So

$$\left\| \mathrm{D}^\alpha(u) \right\|_{L^\infty(\Omega')} \leq C \| u \|_{L^1(\Omega)}$$

Theorem 1.11 (uniqueness of solutions for the Dirichlet problem). Suppose Ω is bounded, and $u_1, u_2 \in C^2(\Omega) \cap C^0(\overline{\Omega})$ with

$$\begin{cases} \Delta u_1 = \Delta u_2 & \text{in } \Omega \\ u_1 = u_2 & \text{on } \partial\Omega \end{cases}$$

Then $u_1 = u_2$ in $\overline{\Omega}$.

[2] This follows from the fact that Ω' is relatively compact

Proof. Set $w = u_2 - u_2$. Then w is harmonic function in Ω, and $w = 0$ on $\partial\Omega$. By the weak maximum principle, we get that $w = 0$ in Ω. □

Remark 1.12. Of course, we can integrate by parts to get the same result, but the weak maximum principle will apply for non divergence form equations.

Lecture 3

Theorem 1.13 (Liouville). If $u \in C^\infty(\mathbb{R}^n)$ is harmonic, and grows sublinearly at infinity. Then u is constant.

Remark 1.14. "growing sublinearly" means that

$$|u(x)| \leq C\left(1 + |x|^\alpha\right)$$

where $0 < \alpha < 1$.

Proof. From derivative estimates, we know that for all $x \in \mathbb{R}^n$, we have that

$$|Du(y)| \leq \frac{C}{\rho} \sup_{B_\rho(y)} |u|$$

Plugging in the growth assumption, we get that

$$|Du(y)| \leq \frac{C}{\rho}(1 + (\rho + |y|)^\alpha)$$

Taking $\rho \to \infty$, we get that $Du(y) = 0$. But y was arbitrary, and so we are done. □

1.2 Existence theory for harmonic functions

We will consider the classical problem of solving the Dirichlet problem for the Laplacian on bounded domains $\Omega \subseteq \mathbb{R}^n$, and $\varphi : \overline{\Omega} \to \mathbb{R}$ continuous. We wish to find $u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$ such that

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega \end{cases}$$

We will assume for simplicity that $\partial\Omega$ is smooth, and $\varphi \in C^\infty(\Omega)$.

We have (at least) three methods to solve this problem.

1. Hilbert space methods (c.f. Analysis of PDEs). Use the Riesz representation theorem to obtain a solution $u \in H^1(\Omega)$, and deal with regularity afterwards. This relies on the equation being linear.
2. Direct method of calculus of variations. We can rephrase $\Delta u = 0$ as a variational problem. That is, the Euler-Lagrange equation of
$$\int |Du|^2$$
and prove existence using the functional.
3. Perron's method. Use the fact that solvability in balls impies solvability in more general domains. This method is based on maximum principles.

Remark 1.15. In all cases, we obtain a rougher solution first, and improve regulaity later.

We will focus on the second method. Define

$$\mathscr{S} = \left\{ w \in \mathrm{H}^1(\Omega) \mid w - \varphi \in \mathrm{H}^1_0(\Omega) \right\}$$

That is, H^1 functions which agree with φ on the boundary. Clearly $\varphi \in \mathscr{S}$, and so $\mathscr{S}$ is non-empty. Set

$$\mathcal{E}[w] = \int_\Omega |\mathrm{D}w|^2$$

and define

$$\beta = \inf_{w \in \mathscr{S}} \mathcal{E}[w]$$

By the definition of the infimum, there exists a seqeunce $(w_j) \subseteq \mathscr{S}$, such that

$$\mathcal{E}[w_j] \to \beta$$

We want to extract a convergent subsequence and show that its limit is a solution. Clearly for j large, we have that

$$\int_\Omega \left|\mathrm{D}w_j\right|^2 \leq \beta + 1$$

Since $w_j - \varphi \in \mathrm{H}^1_0(\Omega)$, by the Poincaré inequality,

$$\int_\Omega \left|w_j - \varphi\right|^2 \leq C \int_\Omega \left|\mathrm{D}(w_j - \varphi)\right|^2$$

In particular, this implies that

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta) < \infty$$

Indeed,

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C\left\|\mathrm{D}w_j - \mathrm{D}\varphi\right\|^2_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta)$$

Expanding the left hand side, we get

$$\left\|w_j\right\|^2_{L^2(\Omega)} - 2\left\langle w_j, \varphi \right\rangle_{L^2(\Omega)} \leq C(\Omega, \varphi, \beta)$$

In particular, by Young's inequality,

$$\left\|w_j\right\|^2_{L^2(\Omega)} \leq C(\varphi, \Omega, \beta) + \varepsilon\left\|w_j\right\|^2_{L^2(\Omega)} + \frac{1}{\varepsilon}\left\|\varphi\right\|^2_{L^2(\Omega)}$$

Take $0 < \varepsilon < 1$, and rearrange.

So we have that $\left\|w_j\right\|^2_{\mathrm{H}^1(\Omega)} \leq C$, so by Banach-Alaoglu,

$$w_{j_k} \rightharpoonup w \text{ in } \mathrm{H}^1(\Omega)$$

By Rellich-Kondrachov[3],

$$w_{j_k} \to w \text{ in } L^2(\Omega)$$

for some $w \in \mathrm{H}^1(\Omega)$. Hence for all $v \in \mathrm{H}^1(\Omega)$, we have that

$$\int_\Omega \mathrm{D}w_j \cdot \mathrm{D}v \to \int_\Omega \mathrm{D}w \cdot \mathrm{D}v$$

[3]Recall

Theorem 1.16 (Rellich-Kondrachov). Let Ω be bounded, $1 \leq p < n$, then

$$W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$$

and

$$W^{1,p}(\Omega) \Subset L^q(\Omega)$$

for $1 \leq q < p^*$. Here, p^* is the conjugate

$$p^* = \frac{np}{n-p}$$

In particular, when $p = 2$, $p^* > 2$ if $n > 2$.

Also, clearly we have that $w_{j_k} - \varphi \rightharpoonup w - \varphi$ in $H^1(\Omega)$, as φ is smooth. But $w_{j_k} - \varphi \in H^1_0(\Omega)$, but $H^1_0(\Omega) \subseteq H^1(\Omega)$ is norm closed, so it is weakly closed[4]. Hence $w - \varphi \in H^1_0(\Omega)$, so $w \in \mathscr{S}$. Finally, since $\mathcal{E}[\cdot]$ is *sequentially weakly lower semicontinuous*[5] in $H^1(\Omega)$, we have that

$$\mathcal{E}[w] \leq \liminf \mathcal{E}[w_{j_k}] = \beta$$

Hence $\mathcal{E}[w] = \beta$. We have found a global minimum w. That is, for all $v \in H^1_0(\Omega)$, $w + tv \in \mathscr{S}$,

$$f(t) = \mathcal{E}[w + tv] \geq \mathcal{E}[w]$$

Thus, $f'(0) = 0$. But

$$\begin{aligned} f'(0) &= D\mathcal{E}[w](v) \\ &= \lim_{t \to 0} \frac{\mathcal{E}[w + tv] - \mathcal{E}[w]}{t} \\ &= 2 \int_\Omega Dw \cdot Dv \end{aligned}$$

In particular,

$$\int_\Omega Dw \cdot Dv = 0$$

is the weak formulation of $\Delta w = 0$. We will next upgrade the regularity, and show that weak solutions are in fact smooth.

[4] This follows from Hahn-Banach. In fact, it follows for any convex subset of a Banach space.

Lemma 1.17. Let X be a Banach space, for $C \subseteq X$ convex, C is norm-closed if and only if C is weakly closed.

Proof. Weak closed implies norm closed is clear. For the converse, we can show that $X \setminus C$ is weakly open. Let $x_0 \in X \setminus C$, by the Hahn-Banach separation, there exists $\phi \in X'$ such that

$$\varphi|_C = 0 \quad \text{and} \quad \phi(x_0) \neq 0$$

Then

$$\left\{ x \in X \;\middle|\; |\phi(x)| > \frac{1}{2}|\phi(x_0)| \right\} \subseteq X \setminus C$$

is a weakly open neighbourhood of x_0. □

[5] That is, if $u_j \rightharpoonup u$ in $H^1(\Omega)$, then

$$\mathcal{E}[u] \leq \liminf \mathcal{E}[u_j]$$

To see this, note that

$$\int_\Omega Du_j \cdot Dv \to \int_\Omega Du \cdot Dv$$

Setting $u = v$, we see that

$$\int_\Omega Du_j \cdot Du \to \int_\Omega |Du|^2$$

Thus,

$$\begin{aligned} \mathcal{E}[u] &= \lim \int_\Omega Du_j \cdot Du \\ &= \liminf \int_\Omega Du_j \cdot Du \\ &\leq \liminf \mathcal{E}[u_j]^{1/2} \mathcal{E}[u]^{1/2} \end{aligned}$$